

Contents

Lesson 3: Variables	1
1. Variables → Python Variables	1
2. Naming Variables	1
Rules for Naming Variables	2
3. Reassigning Variables	2
4. Using Multiple Variables Together	2
Summary	2

Lesson 3: Variables

In Lesson 2, we learned how Python represents numbers and performs arithmetic. But so far, every calculation we’ve done has used numbers directly — we haven’t stored anything. In this lesson, we introduce **variables**, one of the most fundamental ideas in both mathematics and programming.

1. Variables → Python Variables

In mathematics, we use letters like x , y , or n to represent unknown or changing values. For example, in the equation

$$x = 5$$

we say “ x is equal to 5” — x is a placeholder that holds the value 5.

Python works the same way. We can store a value in a variable using the `=` sign, called the **assignment operator**.

```
x = 5
```

This line does not check whether `x` equals 5 — it *assigns* the value 5 to `x`. This is an important distinction from mathematics, where $x = 5$ is usually a statement of equality. In Python, `=` means “store this value in this name,” not “these two things are equal.”

Once a variable is assigned, we can use it just like a number.

```
x = 5
x + 3
```

This gives 8, since Python substitutes the value stored in `x` wherever `x` appears — exactly like substituting a value into an algebraic expression.

2. Naming Variables

In mathematics, we’re used to short, single-letter names like x , y , n , or θ . Python allows short names too, but also allows full descriptive words, which becomes useful as our programs grow.

```
radius = 7
area = 3.14159 * radius ** 2
area
```

Here, `radius` and `area` are variables with meaningful names, making the code easier to read — this connects directly to the formula for the area of a circle:

$$A = \pi r^2$$

Notice how closely the Python line `area = 3.14159 * radius ** 2` mirrors the mathematical formula itself — this is the heart of what it means to use Python as a language of mathematics.

Rules for Naming Variables

Python has a few strict rules for what a variable name can be:

- It must start with a letter or an underscore (not a number).
- It can only contain letters, numbers, and underscores — no spaces or symbols.
- It is case-sensitive, just like everything else in Python (`Area` and `area` are different variables).

For example, `2x` is **not** a valid variable name in Python, even though it's common notation in mathematics. We would need to write it as `two_x` or simply reorder our thinking: use `x = 2` and compute `2 * x` instead.

3. Reassigning Variables

Unlike a fixed mathematical constant, a variable's value in Python can change as the program runs.

```
x = 5
x = x + 1
x
```

Here, Python first calculates `x + 1` using the *current* value of `x` (5), giving 6, and then stores that new value back into `x`. After this line runs, `x` is no longer 5 — it's 6.

This is a subtle but important shift from algebra: in mathematics, $x = x + 1$ has no solution (no number equals itself plus one). But in Python, this is a perfectly valid instruction meaning “update `x` to be one more than it currently is.” This idea — updating a variable's value — is the same *state and update rule* concept we'll rely on heavily once we return to loops.

4. Using Multiple Variables Together

Just as a mathematical formula often involves several variables, Python lets us combine multiple variables in a single expression.

```
base = 10
height = 4
area_of_triangle = 0.5 * base * height
area_of_triangle
```

This mirrors the familiar formula

$$A = \frac{1}{2}bh$$

giving us 20.0 — and just like before, notice how directly the Python line reflects the mathematical formula it's based on.

Summary

Mathematical idea	Python
$x = 5$ (a placeholder for a value)	<code>x = 5</code> (assigns 5 to <code>x</code>)
Substituting a value into an expression	Using a variable in a calculation
A formula with multiple variables, e.g. $A = \pi r^2$	A Python expression using multiple variables

A variable is a name that holds a value — and just like in mathematics, we can use that name in place of the value itself.

Dr. Aselebe Lateefat Olanike